

III. Biology

Subdivisions of Biology

Notes

- The word 'Science' comes from the Latin word 'scientia', meaning 'knowledge'.
- Science is the systematic knowledge of the physical or material world gained through observation and experimentation.
- Science has been classified into – (1) Natural Science and (2) Social Science.
- Natural Science has further classified into – (i) Physical Science and (ii) biological Science which has been further divided into many branches and sub-branches.
- **Lamarck** and **Treviranus** introduced the term '**biology**' independently in modern sense in 1802.

Branches and Subbranches of Biology

Zoology – Detailed study of different living and extinct animals.

Botany – Detailed study of plants and their life.

Palaeontology – The study of history of life on Earth as based on fossils. Fossils are the remains of plants, animals, fungi, bacteria and single celled living things that have been found in layers of rock materials or impressions of organisms preserved in rocks.

Palaeobotany – Study of plant fossils.

Ecology – The branch of biology that deals with the relation of organisms to one another and to their physical surroundings.

Genetics – The study of heredity or how the characteristics of living things are transmitted from one generation to the next. Every living thing contains the genetic material that makes up DNA molecule. This material is passed on when organism reproduces. The basic unit of heredity is '**Gene**'.

Physiology – Study of normal functions of living creatures and their parts.

Pedology – It is the study of soils in their natural environment. It deals with pedogenesis, soil morphology and soil classification.

Edaphology – It concerns with the influence of soils on living things, particularly plants.

Gerontology – Study of the social, cultural, physiological, cognitive, and biological aspects of ageing, or the scientific study of old age, the process of ageing and the particular problems of old people.

Ethology – It is the scientific and objective study of non-human animal behaviour usually with a focus on behaviour under natural conditions, and viewing behaviour as an evolutionary adaptive trait.

Bionics – It is the study of mechanical systems that function like living organisms or parts of living organisms. Bionics is the application of biological methods and systems found in nature to the study and design of engineering systems and modern technology. The word 'bionic' was coined by Jack E. Steele in 1958.

Bionomics – The study of the mode of life of organisms in their natural habitat and their adaptations to their surroundings. Bionomics is the comprehensive study of an organism and its relation to its environment. It is also referred as 'ecology'.

Bionomy – Branch of biology concerned with the laws of life.

Biometry – It is also known as biostatistics. It is the development and application of statistical and mathematical methods to analyse the data results from biological observations and phenomenon.

Phrenology – The detailed study of the shape and size of the cranium as a supposed indication of character and mental abilities.

Anthology – Study of flowers.

Agrostology – It deals with the scientific study of true grasses. It is also called **Graminology** – Scientific study of grasses (the family Poaceae or Gramineae).

Palynology – The study of pollen grains and other spores especially as found in archaeological or geological deposits. Pollen extracted from such deposits may be used for radiocarbon dating and for studying past climates and environments by identifying plants then growing.

Oncology – A branch of biology that deals with the prevention, diagnosis and treatment of tumours/cancer.

Teratology – The scientific study of congenital abnormalities and abnormal formations.

Osteology – It is the scientific and detailed study of the structure of bones, skeletal elements, teeth, microbone, morphology etc.

Chondriology – A branch of anatomy that deals with cartilage.

Serology – The scientific study or diagnostic examination of blood serum, especially with regard to the response of the immune system to pathogens or introduced substances.

Pathology – The science of the causes and effects of diseases, especially the branch of medicine that deals with the laboratory examination of samples of body tissue for diagnostic or forensic purposes.

Parasitology – Parasitology is the study of parasites (viruses, bacteria, fungus, worms, insects), their hosts and the relationship between them. Parasites are organisms that use other species of plants and animals as a host. The hosts provide the environment in which the parasite lives.

Virology – Virology is the study of viruses - submicroscopic, parasitic particles of genetic material contained in a protein coat and virus - like agents.

Bacteriology – Study of the morphology, ecology, genetics and biochemistry of bacteria as well as many other aspects related to them.

Entomology – A branch of zoology that deals with insects.

Malacology – Malacology is the branch of invertebrate zoology that deals with the study of the Mollusca, the second -largest phylum of animals in terms of described species after the arthropods. Molluscs include snails and slugs, clams, octopus, and squid (largest invertebrate) and numerous other kinds, many of which have shells made of calcium carbonate.

Ichthyology – Ichthyology, also known as fish science, is the branch of zoology that deals with fishes.

Serpentology – A branch of zoology that studies snakes.

Herpetology – Herpetology is the branch of zoology which deals with the study of reptiles and amphibians such as snakes, turtles, lizards, tortoises, crocodilians etc. It deals with their behaviour, geographic ranges, physiologies, development, genetics etc.

Ornithology – Ornithology is a branch of zoology that deals with the study of birds.

Dermatology – A branch of medicine concerned with the diagnosis and treatment of skin, nails and hair diseases.

Trophology – Study of nutrition. A nutritional approach that advocates specific combinations of foods as central to good health and weight loss.

Cell biology – Cell biology is a branch of biology that studies the structure and function of the cell, which is the basic unit of life.

Histology – The branch of biology dealing with the study of the microscopic structure of tissues.

Embryology – A branch of biology concerned with the study of embryos and their development.

Phycology or Algology – A branch of biology which deals with the scientific study of seaweeds and other algae.

Mycology – A branch of biology which deals with the scientific study of fungi.

Lichenology – It is the study of lichens, symbiotic organisms made up of an intimate symbiotic association of a microscopic alga (or a cyanobacterium) with a filamentous fungus (mycobiont).

Main Terminologies and related by-products

Apiculture – The raising and care of bees for commercial (honey production) and agricultural purposes (cross pollination).

Aviculture – The breeding and rearing of birds.

Aquaculture – The rearing of aquatic animals or the cultivation of aquatic plants for food.

Algaculture – Algaculture is a form of aquaculture involving the farming of species of algae. Algae are used in wastewater treatment facilities, reducing the need for greater amounts of toxic chemicals. Algae can be used to capture fertilizers in runoff from farms, when subsequently harvested, the enriched algae itself can be used as fertilizer. Microalgae are called phytoplanktons.

Pisciculture – The controlled breeding and rearing of fishes.

Agriculture – The science or practice of farming, including cultivation of the soil for the growing of crops and the rearing of animals to provide food, wool and other products.

Sericulture – The production of silk and the rearing of silkworms for this purpose.

Silviculture – Silviculture is the practice of controlling the growth, composition, health and quality of forests to meet diverse needs and values.

Spongiculture – The cultivation and growing of sponges.

Horticulture – Horticulture is the science and art of growing fruits, vegetables, flowers and crops like spices, condiments and other plantation crops.

Floriculture – Floriculture or flower farming, is a discipline of horticulture concerned with the cultivation of flowering and ornamental plants for gardens and for floristry, comprising the floral industry.

Olericulture – Olericulture is the science of vegetables growing, dealing with the culture of non-woody (herbaceous) plants for food.

Arboriculture – Cultivation of trees and shrubs.

Pomology – The science that deals with fruit and fruit growing.

Agronomy – A branch of agriculture dealing with crop production and soil management.

Vermiculture – The cultivation of earthworms, especially in order to use them to convert organic waste into fertilizer.

Ostriculture – The breeding of oysters for food and pearls.

Heliculture – It is also known as snail farming. It is the process of raising land snails especially for human use, either to use their flesh to eat or to obtain snail slime for use in cosmetics and snail eggs for human consumption as a type of caviar.

Viticulture – Cultivation and harvesting of grapes.

Mariculture – Mariculture is a specialized branch of aquaculture involving the cultivation of marine organisms (animal and plant) for food and other products.

Moriculture – Moriculture is the science of mulberry cultivation for silk production because silk worm (*Bombyx mori*) reared on it.

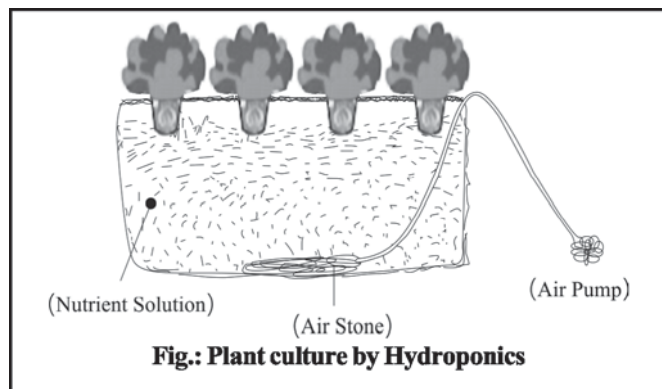
Citriculture – Cultivation of citrus fruit trees.

Vegeculture – The cultivation of vegetables, also the cultivation of plants, especially root crops, propagated by vegetative means.

Aeroponics – A plant-cultivation technique without soil, in which the roots hang suspended in the air while nutrient solution is delivered to them in the form of a fine mist.

Aeroculture – A method of growing plants without soil by suspending them above sprays that constantly moisten the roots with water and nutrients.

Hydroponics – It is a method of growing plants without soil by instead of using mineral nutrient solutions in a water solvent (see fig.).



Question Bank

1. Study of Zoology deals with :

- (a) Living animals only
- (b) Living plants only
- (c) Both living and dead animals
- (d) Both living and dead plants

U.P.P.C.S. (Pre) 1995

Ans. (c)

Zoology is a branch of biology. It deals with the study of both living and extinct animals including their structure, embryology, evolution, classification, habits, and distribution. Aristotle is known as 'Father of Zoology'.

2. Match the following List-I to List-II and by using the given codes in below list. Choose the correct answer.

List -I	List-II
A. Birds	1. Palaeobiology
B. Inheritance	2. Ecology
C. Environment	3. Ornithology
D. Fossil	4. Genetics

Code :

	A	B	C	D
(a)	1	3	4	2
(b)	3	4	2	1
(c)	4	2	1	3
(d)	2	4	1	3

U.P. Lower Sub. (Pre) 1998

Ans. (b)

The correctly matched order is given below :

Study of Birds	—	Ornithology
Study of Inheritance	—	Genetics
Study of Environment	—	Ecology
Study of Plant Fossils	—	Palaeobiology

3. The study of the flower is known as :

- (a) Phrenology
- (b) Anthology
- (c) Agrostology
- (d) Palynology

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

Phrenology	—	Study of human skull
Anthology	—	Study of flowers
Agrostology	—	The study of grasses
Palynology	—	Study of pollen grains and other spores

4. The scientific study of insects is known as :

- (a) Ichthyology (b) Entomology
(c) Parasitology (d) Malacology

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

Ichthyology – Study of fish
Entomology – Scientific study of insects
Parasitology – Study of parasitic organisms
Malacology – Study of molluscs and its shells

5. Which of the following subjects of data set is associated with the study of concerning population and humanity?

- (a) Ecology (b) Genetics
(c) Demography (d) Virology

U.P.P.C.S. (Pre) 1994

U.P.G.I.C2017

Ans. (c)

Ecology - Study of interrelationship between organisms and environment
Genetics - Study of heredity and variation
Demography - Statistical study of populations including human beings
Virology - Study of viruses

6. The comparative study of the different cultures and the scientific explanation is known as–

- (a) Ethnology (b) Ethnography
(c) Ethology (d) Ethics

R.A.S./R.T.S. (Pre) 1994

Ans. (a)

Ethnology – Science dealing with different races of mankind.
Ethnography – A study of specific culture.
Ethology – Study of the animal behaviour.
Ethics – Study of ethical duties.

7. The study of properties and method of biological systems found in nature and use of this knowledge in mechanical world is known as–

- (a) Bionics (b) Bionomics
(c) Bionomy (d) Biometry

R.A.S./R.T.S. (Pre) 1994

Ans. (a)

Bionics – Study of properties and method of biological systems found in nature and use of this knowledge in mechanical world.
Bionomics – Comprehensive study of an organism and its relation to its environment.
Bionomy – Study of laws of life
Biometry – Statistical study of biological problem.

8. Under Palaeobotany, one studies –

- (a) Fossil of animals (b) Algae
(c) Fungi (d) None of these

Uttarakhand P.C.S. (Pre) 2005

Ans. (d)

Palaeobotany is the branch of palaeontology or palaeobiology dealing with the recovery and identification of plant remains from geological contexts and their use for the biological reconstruction of past environments. In India, Prof. Birbal Sahni is known as Father of Indian Palaeobotany.

9. Rearing silk worms is known as –

- (a) Apiculture (b) Horticulture
(c) Sericulture (d) Floriculture

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (c)

Apiculture – Rearing bees
Horticulture – Study of fruit and flower producing plants
Sericulture – Rearing silk worm
Floriculture – Study of flowers for decoration work

10. Honey bees are used in :

- (a) Sericulture (b) Tissue culture
(c) Apiculture (d) Pisciculture

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

Honey bees are used in apiculture.

11. Which one of the following is not correctly matched?

- | Cultivation | Nomenclature of Cultivation |
|---------------------------|-----------------------------|
| (a) Flower Cultivation | – Floriculture |
| (b) Crop Cultivation | – Agronomy |
| (c) Vegetable Cultivation | – Horticulture |
| (d) Fruit's Cultivation | – Pomology |

U.P. Lower Sub. (Pre) 2015

Ans. (c)

Horticulture is a special branch of agriculture. Pomology, olericulture, and floriculture come under this branch. The name floriculture and pomology used in option (b) and (d) are of specific terminology. In option (c) the word horticulture has been used for the cultivation of vegetables. The specific terminology for cultivation of vegetables should be 'Olericulture'. If all the names used in the given options are considered in literal sense then option (c) is not correctly matched because the word olericulture should be used in place of horticulture. The word crop has a wide ranged vocabulary containing all types of crops like eatable, non-eatable and animal foods etc. Thus the word 'agronomy' correctly matched for crop cultivation.

12. The study related to the plants being used as vegetable is called –

- (a) Floriculture (b) Pomology
(c) Horticulture (d) Olericulture

Uttarakhand P.C.S. (Pre) 2010

Ans. (d)

Olericulture is the science of vegetable growing, dealing with the culture of non-woody (herbaceous) plants for food.

13. Which of the following is not correctly matched?

- (a) Apiculture – Honey bee
(b) Sericulture – Silkworm
(c) Pisciculture – Insect of lakh
(d) Horticulture – Flower

U.P. Lower Sub. (Pre) 2004

Ans. (c)

Pisciculture is related to the fish hatchery. Other options are correctly matched.

14. Which of the following pairs is correctly matched? (word and meaning)

- (a) Lexicography – Compilation of dictionary
(b) Philology – Collection of postal stamps
(c) Entomology – Study of human body
(d) Iconography – Study of insects

Chhattisgarh P.C.S. (Pre) 2008

Ans. (a)

Lexicography – Compilation of dictionary
Philology – Study of language structure and its history
Entomology – Study of insects
Iconography – Study of idols/arts etc.

15. Lexicography is related to :

- (a) Structure of human body
(b) Compilation of dictionary
(c) Secret of handwriting
(d) Teaching of pictures & models.

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

Lexicography is related to a glossary of language design, compilation, and evaluation of general dictionaries.

16. Which one of the following is produced through viticulture :

- (a) Silk (b) Earthworm
(c) Honey (d) Grapes

U.P.P.C.S. (Mains) 2003

Ans. (d)

Viticulture is the science, production, and study of grapes. It deals with the series of events that occur in the vineyard.

17. What is Vermiculture?

- (a) The science of raising worms
(b) The science of studying animals
(c) The science of studying fishes
(d) The science of killing worms

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Vermiculture is the process of managing and cultivating earthworms. Earthworms can help turning organic waste into nutrient-rich soil for your garden.

18. In Vermiculture, the worm used is-

- (a) Tapeworm (b) Silkworm
(c) Threadworm (d) Earthworm

U.P.P.C.S. (Mains) 2005

Ans. (d)

See the explanation of above question.

19. Genetics deals with-

- (a) Mendel's laws (b) Organic evolution
(c) DNA structure (d) Heredity and variations

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Genetics is the study of genes, heredity and genetic variations in living organisms.

20. The study of Gerontology, is related to which of the following?

- (a) Child infant (b) Female
(c) Disease of skin (d) Old age

M.P.P.C.S. (Pre) 1993

Ans. (d)

The scientific study of old age, the process of ageing and the particular problems of old people is called Gerontology.

21. The branch of study dealing with old age and ageing is called

- (a) Oncology (b) Gerontology
(c) Teratology (d) Ornithology

U.P. Lower Sub. (Pre) 2008

Ans. (b)

Gerontology is the study of the social, psychological, cognitive and biological aspects of ageing. Oncology is the branch of medical science that deals in treating people suffering from cancer. Teratology is the study of abnormalities of physiological development. Ornithology is a branch of zoology that is related to the study of birds.